THESIS RECAP

OCTOBER 2026

Introduction

Setting

Given a sample space Ω, let inputs and labels be generated by $X : \Omega \to \mathcal{X} = \{\bar{x}_1, \dots, \bar{x}_N\}$ and $Y : \Omega \to \mathcal{Y} = \{-1, 1\}$. Let p be a probability distribution on $\mathcal{X} \times \mathcal{Y}$, with

$$p_{j,y} = \Pr[X = \bar{x}_j, Y = y], \qquad p_{j,y} \geq 0, \qquad \sum_{j=1}^{N} \sum_{y \in \mathcal{Y}} p_{j,y} = 1.$$

Let $D = (X_i, Y_i)_{i=1}^n$ be a dataset of n indipendent copies of (X, Y) with common distribution p. We indicate the realizations of D with lower-case x and y. Let $\mathcal{H} \subseteq \{h : \mathcal{X} \to \mathcal{Y}\}$ be a nonempty, finite set of classifiers. Given a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}^+$, define the population and empirical risks by

$$R(h) = \mathbb{E}_D(\ell(h(X), Y)) = \sum_{\bar{x} \in \mathcal{X}} \sum_{\bar{y} \in \{-1,1\}} \ell(h(\bar{x}), \bar{y}) p(\bar{x}, \bar{y}).$$

and

$$\hat{R}_D(h) = \frac{1}{n} \sum_{i=1}^{n} \ell(h(X_i), Y_i).$$

We select the classifier that minimizes the empirical risk (ERM), i.e.

$$\hat{h}_D \in \arg\min_{h \in \mathcal{H}} \hat{R}_D(h)$$

which, importantly, is a random variable of the data D. Notice that at least one such $\hat{h}_D$ exists, since $\mathcal{H}$ is finite and nonempty.

Specifically, in this thesis

- ℓ is the counting loss $\ell(h(x), y) = \mathbf{1}(h(x) \neq y) = \frac{1 - h(x)y}{2}$;
- classifiers $h : \mathcal{X} \to \mathcal{Y}$ are binary decision trees;
- given a depth d, $\mathcal{H}_d$ is the set of binary decision trees of depth at most d, and we consider a number of such depths.

Research question

In machine learning, there exists a large number of algorithms, notably neural networks trained with gradient descent, which achieve great performances at prediction tasks on unseen data. A big question in learning theory is to understand the relationship between the behavior of such algorithms on the available data and their future behavior on unseen data (generalization). How can we predict how this future behavior will look like?

A number of results have tried to answer this question. Recent works seem to suggest that the classical theory fails to explain generalization, in particular they fail to describe the behavior of the test error of learning algorithms as the comlpexity of the hypotheses set increases. This thesis seeks to understand whether this is true by testing theoretical results against empirical evidences. We study the setting of a finite hypotheses set of binary decision trees. We look for a function that reproduces qualitatively the test error as the maximum depth of the decision trees (i.e. the complexity of the hypotheses space) increases. The contribution of identifying and empirically validating such a function is two-fold. On the one hand, it explains the shape of the test error curve of decision trees; on the other hand, it may allow to implement the same framework and techniques to study the generalization performance of more complex algorithms, like neural networks trained with gradient descent.

In particular, we study bounds on the expcted gap between the test and train error of the ERM,

$$\mathbb{E}_D\left[(R - \hat{R}_D)(\hat{h}_D)\right].$$

Notice that since $\hat{h}_D$ and $\hat{R}_D$ depend on the same dataset D, $\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \neq R(\hat{h}_D)$, i.e. the empirical risk is a biased estimate of the true risk. A proof of this fact is found in Appendix B.

We look for a bound that reproduces the shape of the empirical test error curve qualitatively. In particular, we want this function to have the same asymptotic behavior as $\mathbb{E}_D[R(\hat{h}_D)]$, as well as to be sensitive to changes in the structure of the learning problem which affect the test error curve.

All proofs are in Appendix B.

1. Union bound over $\mathcal{H}$

In this section, we study the first bound on the quantity $\mathbb{E}_D[(R - \hat{R}_D)(\hat{h}_D)]$, to which we refer as the *gap* for the rest of this work, without specifying that it is computed at $\hat{h}_D$ everytime. This result is obtained starting from a concentration result (Hoeffding's inequality, see Theorem **??** in the Appendix) on the random variable $\hat{R}_D(h)$ for a fixed function $h \in \mathcal{H}$. This is extended to a result on $\hat{h}_D$, which depends on the data D, by taking a union bound (see Proposition **??** in the Appendix) on the probability that the event holds for any function $h \in \mathcal{H}$.

Theorem 1.1.

$$\mathbb{E}_D[R(\hat{h}_D) - \hat{R}_D(\hat{h}_D)] \;\leq\; O\left(\sqrt{\frac{\log|\mathcal{H}|}{n}}\right) \tag{1}$$

We plot the bound (1) to compare it with the empirical behavior of the test error in Figure 1.

Experimental design. We create the support $\mathcal{X}$ as a discretization over N numbers of the interval $[0, 10]$. We sample from it with the uniform measure $p_{\mathcal{X}}(\bar{x}) = \sum_{\bar{y}\in\mathcal{Y}} p(\bar{x}, \bar{y}) = \frac{1}{N}$. We generate the labels as

$$y \leftarrow \text{sign}(h^*(x)) \cdot b$$

where $h^* : \mathcal{X} \to \mathcal{Y}$ and $b \sim \text{Bernoulli}(p)$, $p \in (0, 1)$, that is the noise flipping the labels with probability $1 - p$. Our first choice of h^* is a parabola centered horizontally at 5, the midpoint of the support $\mathcal{X}$, and vertically so that roughly half of the points fall above the x-axis and half below. Then, for the sake of flexibility, we pick a more complex function, a sinusoid. We maintain this choice throughout our experiments. Hence,

$$h^* : x \mapsto 5\,\sin\left(\frac{\pi\,(m+1)}{10}\,x\right)$$

where m is the number of times h^* crosses the x-axis in the open interval $(0, 10)$, yielding $m+1$ contiguous blocks of $1s$ and $-1s$. Also, we test a Gaussian noise instead of a Bernoullian, concluding that there is no interesting difference in the performance of the algorithm as long as the portion of misclassified inputs due to the noise is the same. Hence, we stick to $\varepsilon \sim \text{Bernoulli}(p)$. All in all, our law is

$$y \leftarrow \text{sign}\left(5 \sin\left(\frac{\pi\,(m+1)}{10}\,x\right)\right) \cdot b.$$

Default values are $p = 0.8$ and $m = 10$. Notice also that our $\text{sign}(\cdot)$ maps 0 to 1. Plots of the afore-mentioned preliminary experiments are in Appendix A.

We proceed as follows. We generate a test set with size $n_{\text{test}} = 200,000$ sampling from $\mathcal{X}$ with $p_{\mathcal{X}}$ and generating the labels as above. Then, for each depth $d \in [1, 7\lceil \log_2 n \rceil] \subset \mathbb{N}$ we do the following. For a number $T = 100$ of iterations, we

- generate a train set ($n = 50$);
- fit a decision tree with depth at most d (DecisionTreeClassifier algorithm from the scikit-learn library);
- compute the train error, the test error, and the differene between the two (the gap).

Then, we average the resulting quantities over the T samples, obtaining empirical estimates of the expected values of the train error, test error, and gap over the disorder of the data.

To plot the theoretical bound, instead, we need to express $|\mathcal{H}_d|$ as a function of d. The exact computation gives

$$|\mathcal{H}_d| = 2 \sum_{k=0}^{\min N-1, 2^d-1} \binom{N-1}{k}$$

Disclaimer: the plots below have been obtained with $|\mathcal{H}_d| \leq 2^d$ which is incorrect, I need to fix it. I tried the formula above or something like $\min(2^N, N^{2^d} 2^{2^d})$ but dependence on d is awful.

We plot, as the maximum depth d increases

(1) the train and the test error curves;

(2) the gap and the theoretical bound $\sqrt{\log |\mathcal{H}|/n}$;

(3) the test error and the sum of train error and the bound (moving $\hat{R}_D(\hat{h}_D)$ to the right).

Observations:

- ▷ the bound is correct,
- ▷ it diverges when instead the empirical test error saturates.

Since the bound naively depends on the cardinality of $\mathcal{H}_d$, which increases with the depth, it diverges and it fails to predict the saturation of the test error. In the following, we consider a different counting technique that accounts for the finite nature of our problem and the structure of our learning algorithm.

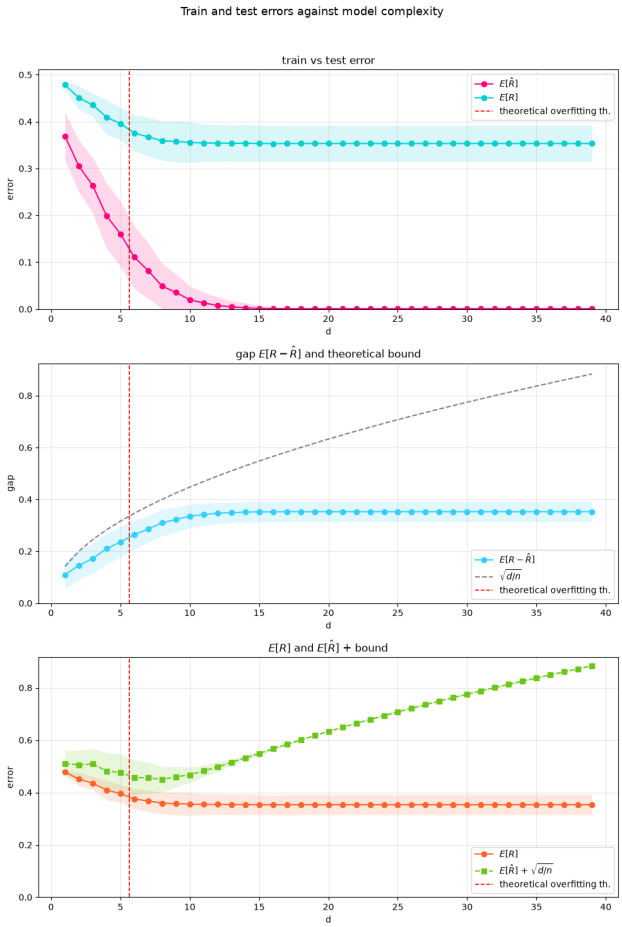


FIGURE 1. $n = 50, N = 10,000, n_{\text{test}} = 200,000, \sigma = 5$, h^* = sine with 10 sign changes

2. Rademacher Complexity and Massart Lemma

Definition 2.1 (Rademacher complexity)**.**

$$\mathcal{R}_n(\mathcal{L}) = \mathbb{E}_{\xi,D}\left[\sup_{\ell_h \in \mathcal{L}} \frac{1}{n}\sum_{i=1}^{n} \xi_i \ell(h(X_i), Y_i)\right]$$

Theorem 2.2 (Rademacher complexity bound)**.**

$$\mathbb{E}_D[(R - \hat{R})(\hat{h})] \leq 2\mathcal{R}_n(\mathcal{L})$$

Lemma 2.3 (Massart)**.**

$$\mathcal{R}_n(\mathcal{L}) \leq \mathbb{E}_D\left[\sqrt{\frac{2\log N_D(\mathcal{H})}{n}}\right]$$

where $N_D(\mathcal{H}) = |\{(\ell_h(X_1, Y_1), \ldots, \ell_h(X_n, Y_n)) : h \in \mathcal{H}\}|$.

Lemma 2.4 (Counting inequalities)**.**

$$N_D(\mathcal{H}) \leq \min\{|\mathcal{H}|, 2^n\}.$$

Putting together this results, we obtain the final bound.

Theorem 2.5 (Rademacher and Massart bound)**.**

$$\begin{aligned}\mathbb{E}_D[(R-\hat{R})(\hat{h}_d)] &\leq 2\mathbb{E}_D\left[\sqrt{\frac{2\log N_D(\mathcal{H})}{n}}\right] \\ &\leq 2\sqrt{\frac{2\min\{\log|\mathcal{H}_d|, \log 2^n\}}{n}} \\ &\leq \min\left\{O\left(\sqrt{\frac{\log|\mathcal{H}_d|}{n}}\right), O(1)\right\} \end{aligned} \tag{2}$$

We plot the bound (2), with as constant term 0.5, in Figures 2 and 3. The experimental design is the same as before.

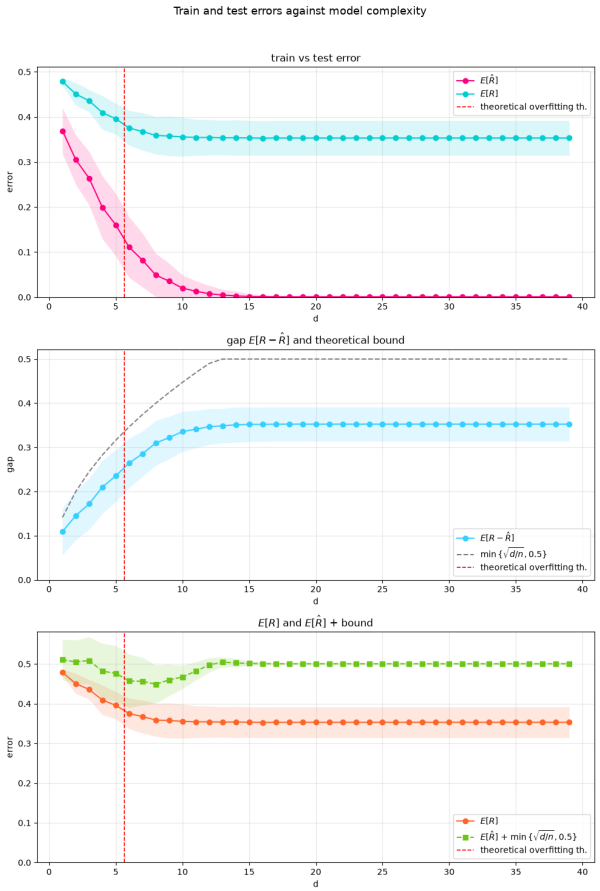


FIGURE 2. Identical parameters as Fig.1

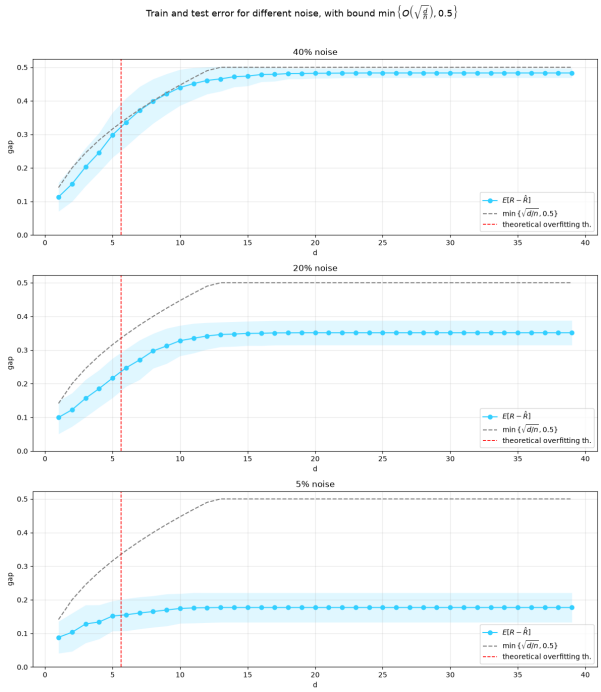


FIGURE 3. Different noise levels

Observations:

- ▷ the bound saturates,
- ▷ the bound does not change for different noise levels.

On one hand, bound (2) saturates mirroring the behavior of the test error curve. On the other hand, the bound and the saturating level are not sensitive to changes in the noise level, as Figure 3 shows. Indeed, the bound saturates because of Lemma 2.4, which bounds the number of loss vectors $N(D)$ with the minimum between the number of classifiers in $\mathcal{H}$, increasing in d, and 2^n, a constant. Hence, at some point the curve does not increase with depth d anymore. This reflects the fact that the Rademacher complexity is naturally bounded by 0.5 in our setting. At the same time, the Rademacher complexity is blind to the structure of our problem, specifically to the level of noise.

In the next section we implement a different theoretical viewpoint to look for a bound that better captures the structure of the learning problem.

3. Dependence on the Best Risk

Set

$$R^*_{\mathcal{H}} = \min_{h \in \mathcal{H}} R(h).$$

We aim at obtaining a bound on the gap at $\hat{h}_D$ in terms of the best risk $R^*_{\mathcal{H}}$. This is because the best risk can be decomposed as

$$R^*_{\mathcal{H}} = \underbrace{R^*}_{\text{Bayes risk}} + \underbrace{R^*_{\mathcal{H}} - R^*}_{\text{Approximation error}} =: \sigma^2 + A_{\mathcal{H}}$$

where

- $\sigma^2 = R^* := \inf_{h: \mathcal{X} \to \mathcal{Y}}$ measures the irreducible classification error due to noise, that we make no matter the hypotheses set we choose;
- $A_{\mathcal{H}}$ measures the excess error we make given that we consider a restricted set of classifiers, $\mathcal{H}$.

So, we hope to obtain a noise and structure-sensitive bound.

Theorem 3.1 (Bound with empirical risk)**.** *With probability at least* $1 - \delta$*,*

$$R(\hat{h}) - \hat{R}(\hat{h}) \leq \sqrt{\frac{\hat{R}(\hat{h}) \log(2N_{2n}(\mathcal{H})/\delta)}{n}} + 16 \frac{\log(2N_{2n}(\mathcal{H})/\delta)}{n}$$

where $N_n(\mathcal{H}) := \sup_{(x_1, y_1), \ldots, (x_n, y_n) \in \mathcal{X} \times \mathcal{Y}} N_D(\mathcal{H})$*.*

Corollary 3.2 (Bound with best risk)**.** *With probability at least* $1 - \delta$*,*

$$R(\hat{h}) - \hat{R}(\hat{h}) \leq \sqrt{\frac{R^*_{\mathcal{H}} \log(4N_{2n}(\mathcal{H})/\delta)}{n}} + 20 \frac{\log(4N_{2n}(\mathcal{H})/\delta)}{n}$$

Corollary 3.3 (Noise and approximation error bound (high probability))**.** *With probability* $1 - \delta$*,*

$$R(\hat{h}) - \hat{R}(\hat{h}) \leq \sqrt{\frac{(A_{\mathcal{H}} + \sigma^2) \log(4N_{2n}(\mathcal{H})/\delta)}{n}} + 20 \frac{\log(4N_{2n}(\mathcal{H})/\delta)}{n}$$

Theorem 3.4 (Noise and approximation error bound)**.**

$$\mathbb{E}_D \left[R(\hat{h}) - \hat{R}(\hat{h}) \right] \leq \min \left\{ O\left(\sqrt{\frac{(A_{\mathcal{H}} + \sigma^2) \log |\mathcal{H}|}{n}} + \frac{\log |\mathcal{H}|}{n} \right), O\left(\sqrt{A_{\mathcal{H}} + \sigma^2} + 1 \right) \right\} \tag{3}$$

We plot bound (3) in Figure 4 and Figure 5.

Figure 6 shows the two terms of the minimum in the bound separately. The Bayes risk, the minimum risk over $\mathcal{H}$ and the approximation error (their difference) for different noise levels are shown in Figure 7.

Experimental design: The fundamental design is the same as before. The Bayes risk R^* is

$$R^* = 1 - p$$

for $p \geq 0.5$, and as before we sue $p \in \{0.6, 0.8, 0.95\}$. $R^*_{\mathcal{H}}$ is computed exactly using dynamic programming. We produce labels using our sine function and the Bernoulli error and we cluster the intervals where labels have the same sign, obtaining M contiguous blocks. We create a dp $k \times 2$ matrix that stores the minimum risk for each number of maximum splits k, up to $\min(M-1, 2^d)$ where d is the maximum depth we are considering, when the last assigned label is 1 or -1 (the columns). Then, for each contiguous block we dynamically update the dp matrix summing the minimum risk among all trees with at most k splits. In the end, we select the minimum risk among the tree predicting 1 or -1 in the very last block.

Disclaimer: For the second term in the min, we plot $\sqrt{A_{\mathcal{H}} + \sigma^2}$ directly without the $+1$, although I am not sure this is motivated, since $R^*_{\mathcal{H}} < 1$ as shown in Figure 7.

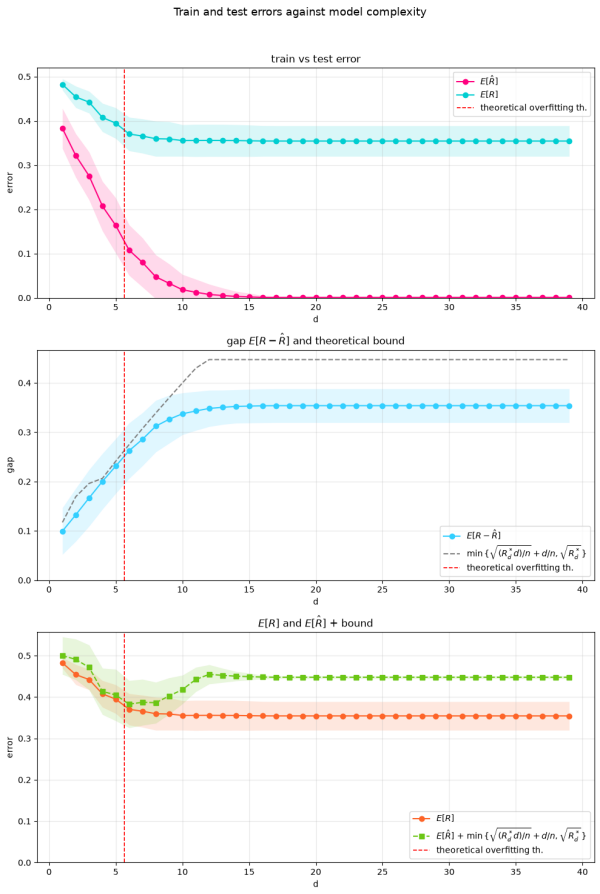


FIGURE 4. Same setting, with Bernoullian noise at 20%

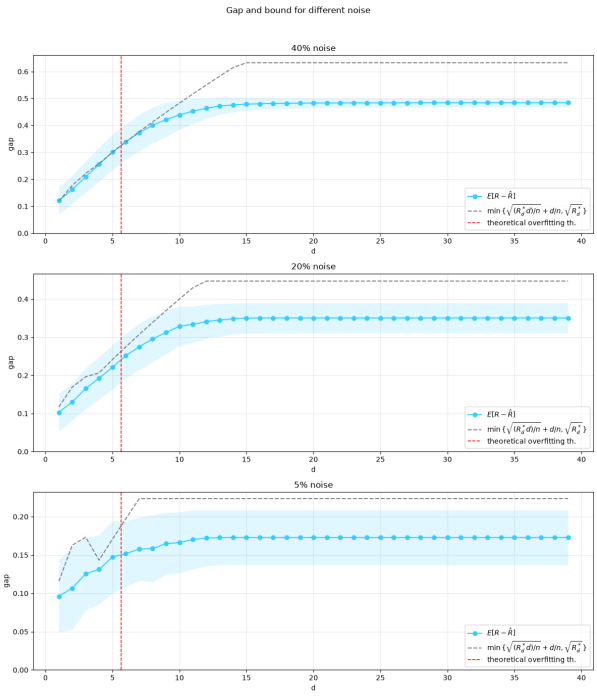


FIGURE 5. Different noise levels

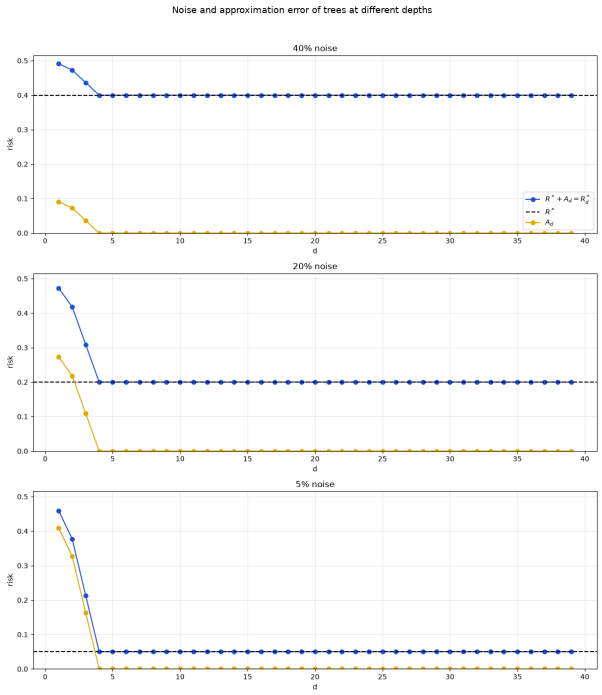


FIGURE 6. The two terms of the minimum in the noise and approx. error bound

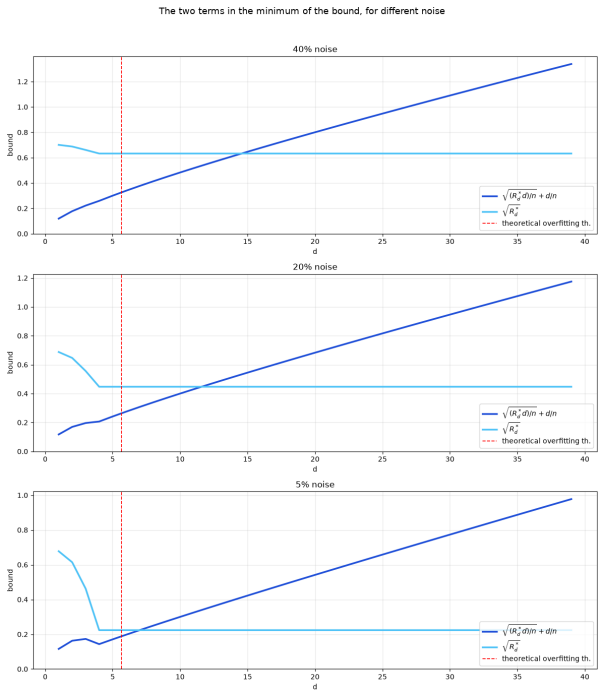


FIGURE 7. Approximation error and Bayes risk

4. Growth function

Definition 4.1 (Growth function)**.** *The growth function of $\mathcal{H}$ is*

$$\Pi_n(\mathcal{H}) := \sup_{x_1,\dots,x_n \in \mathcal{X}} \left|\{(h(x_1),\dots,h(x_n)) : h \in \mathcal{H}\}\right|.$$

The following extends Lemma 2.4.

Lemma 4.2 (Counting inequalities with growth function)**.** *For every sample D of size n,*

$$1 \le N_D(\mathcal{H}) \le \Pi_n(\mathcal{H}) \le \min\{|\mathcal{H}|, 2^n\}.$$

Moreover,

$$\Pi_n(\mathcal{H}) = N_n(\mathcal{H}) = \sup_{(x_1,y_1),\dots,(x_n,y_n) \in \mathcal{X}\times\mathcal{Y}} N_D(\mathcal{H})$$

The relation between N_D and Π_n makes it immediate to obtain growth-function versions of the bounds in Sections 2 and 3 (substitute $N(D) \le \Pi_n$ into Lemma 2.3 and $N_{2n} = \Pi_{2n}$ into Theorem 3.1). Substituting also $\Pi_n(\mathcal{H}) \le \min\{|\mathcal{H}|, 2^n\}$ yields bounds (2) and (3).

5. Going on

The growth function has the additional result that

$$\log \Pi_{\mathcal{H}}(2n) \le d \log \frac{2en}{d}$$

we need to understand if this is useful. Plus, we mentioned the localized Rademacher sometime this summer.

I still need to fix the matter of how to count $|\mathcal{H}|$ in terms of the depth d (the "disclaimer" in Section 1). Setting the constant term at 0.5 in bound (2) is heuristical because I know it works. Needs motivation. Similarly, the disclaimer after bound (3): I say $O(\sqrt{A_{\mathcal{H}} + \sigma^2} + 1) = O(\sqrt{A_{\mathcal{H}} + \sigma^2})$ and plot only $\sqrt{A_{\mathcal{H}} + \sigma^2}$ but it's heuristical as well, and since $A_{\mathcal{H}} + \sigma^2 \leq 1$ probably wrong (however, con il $+1$ non fa nemmeno in tempo a saturare perché l'altro termine che dipende da d rimane più piccolo).

Appendix A. Preliminary experiments

Appendix B. Proofs

Proof of Theorem 1.1. Apply Hoeffding Lemma and a union bound over the hypotheses set $\mathcal{H}$. Tail integrate to obtain a result in expected value. □

Proof of Theorem 2.2. Symmetrization technique. □

Proof of Lemma 2.3. Pass to the supremum over *loss vectors* instead of functions $\ell \circ h$, and bound it with the sum, which makes the $N(D)$ term appear. Then, apply the exponential Hoeffding Lemma and optimize for the coefficient in the exponent. □

Proof of Lemma 2.4. Given a dataset, its realizations through ℓ_h are less than or equal to those through h because applying ℓ cannot increase the number of outputs, and the number of images of the functions in $\mathcal{H}$ are less than or equal to the number of functions $|\mathcal{H}|$. On the other hand, $N_D(\mathcal{H}) \leq 2^n$ because we have at most 2^n n-long sequences of $0s$ and $1s$. □

Proof of Theorem 2.5. Apply Theorem 2.2, Lemma 2.3 and Lemma 2.4. □

Proof of Theorem 3.1. Apply the relative deviation inequality (see Loz's file),

$$\Pr\left[\exists h \in \mathcal{H} : R(h) - \hat{R}(h) > \varepsilon\sqrt{R(h)}\right] \leq 2N_{2n}(\mathcal{H})e^{-2\varepsilon^2/16}$$

with the suitable ε and then isolate $\hat{R}(\hat{h})$ on the RHS. □

Proof of Corollary 3.2. Set $h_0 = \arg\min_{h\in\mathcal{H}} R(h)$, so that $R(h_0) = R^*_{\mathcal{H}}$. Then,

$$\hat{R}(\hat{h}) \leq \hat{R}(h_0). \tag{4}$$

Since h_0 is independent of the data, $\mathbb{E}_D[\hat{R}(h_0)] = R(h_0)$ and hence Bernstein gives, with probability at least $1 - \delta/2$,

$$\hat{R}(h_0) \leq R^*_{\mathcal{H}} + \sqrt{\frac{2R *_{\mathcal{H}} \log(2/\delta)}{n}} + \frac{\log(2/\delta)}{3n}.$$

Since $N_{2n}(\mathcal{H}) \geq 1$ and thus $\frac{\log(2/\delta)}{n} \leq \frac{\log(4N_{2n}(\mathcal{H})/\delta)}{n}$,

$$\hat{R}(h_0) \leq \left(\sqrt{R^*_{\mathcal{H}}} + \sqrt{\frac{4\log(N_{2n}(\mathcal{H})/\delta)}{n}}\right)^2 \tag{5}$$

Substitute (4) into (5), and apply the bound to $\hat{R}(\hat{h})$ in the RHS of Theorem 3.1 at confidence $1-\delta/2$. This yields the desired bound, which holds with probability at least $1-\delta$ by a union bound on the events ((5) and Theorem 3.1) that we require to hold simultaneously. □

Proof of Corollary 3.3. Substitute $R^*_{\mathcal{H}} = \sigma^2 + A_{\mathcal{H}}$ into Corollary 3.2. □

Proof of Theorem 3.4. Let $\text{gap} = (R - \hat{R}(\hat{h}))$ and $C = \sqrt{\frac{\log(4N_n(\mathcal{H})/\delta)}{n}}$. Corollary **??** gives

$$\Pr\left[\text{gap} \leq \sqrt{R^*_{\mathcal{H}}}\, C + 20C^2\right] \geq 1 - \delta$$
$$\implies \Pr\left[\text{gap} > \sqrt{R^*_{\mathcal{H}}}\, C + 20C^2\right] \leq \delta$$

Set $\varepsilon = \sqrt{R^*_{\mathcal{H}}}\, C + 20C^2$. Let us now find $\delta(\varepsilon)$:

$$20C^2 + \sqrt{R^*_{\mathcal{H}}}\, C - \varepsilon = 0 \implies C = \frac{-\sqrt{R^*_{\mathcal{H}}} + \sqrt{R^*_{\mathcal{H}} + 80\varepsilon}}{40}$$

So we have

$$\frac{\log(4N_{2n}(\mathcal{H})/\delta)}{n} = \left(\frac{\sqrt{R^*_{\mathcal{H}} + 80\varepsilon} - \sqrt{R^*_{\mathcal{H}}}}{40}\right)^2$$

$$\frac{4N_{2n}(\mathcal{H})}{\delta} = \exp\left(n\left(\frac{\sqrt{R^*_{\mathcal{H}} + 80\varepsilon} - \sqrt{R^*_{\mathcal{H}}}}{40}\right)^2\right)$$

and hence

$$\delta(\varepsilon) = 4N_{2n}(\mathcal{H}) \exp\left(-\frac{n}{1600}\left(\sqrt{R^*_{\mathcal{H}} + 80\varepsilon} - \sqrt{R^*_{\mathcal{H}}}\right)^2\right).$$

Using that $\mathbb{E}[X] = \int_0^\infty \Pr[X \geq \varepsilon]\, d\varepsilon$ for $X \geq 0$, applied to $\text{gap}^+ \coloneqq \max\{\text{gap}, 0\}$,

$$\mathbb{E}_D[\text{gap}] \leq \mathbb{E}_D[\text{gap}^+] = \int_0^\infty \Pr[\text{gap} \geq \varepsilon]\, d\varepsilon \leq \int_0^\infty \min\{1, \delta(\varepsilon)\}\, d\varepsilon.$$

By a shift of perspective, looking at the integral as a sum of horizontal stripes instead of vertical stripes in the graph of $\varepsilon \mapsto \delta(\varepsilon)$, we obtain

$$\int_0^\infty \min\{1, \delta(\varepsilon)\}\, d\varepsilon = \int_0^1 \delta^{-1}(t)\, dt = \int_0^1 \varepsilon(\delta)\, d\delta$$

Hence, writing $\varepsilon(\delta) = \sqrt{R^*_{\mathcal{H}}}\, C + 20C^2$ explicitly,

$$\mathbb{E}_D[\text{gap}] \leq \int_0^1 \varepsilon(\delta)\, d\delta = \sqrt{\frac{R^*_{\mathcal{H}}}{n}} \int_0^1 \sqrt{\log \frac{4N_{2n}(\mathcal{H})}{\delta}}\, d\delta + \frac{20}{n} \int_0^1 \log \frac{4N_{2n}(\mathcal{H})}{\delta}\, d\delta.$$

The logarithm is the only part depending on δ, and its integral is

$$\int_0^1 \log \frac{4N_{2n}(\mathcal{H})}{\delta}\, d\delta = \log(4N_{2n}(\mathcal{H})) - \int_0^1 \log \delta\, d\delta = \log(4N_{2n}(\mathcal{H})) + 1.$$

Applying Jensen's inequality to the square root term,

$$\int_0^1 \sqrt{\log \frac{4N_{2n}(\mathcal{H})}{\delta}}\, d\delta \leq \sqrt{\int_0^1 \log \frac{4N_{2n}(\mathcal{H})}{\delta}\, d\delta} = \sqrt{\log(4N_{2n}(\mathcal{H})) + 1},$$

we obtain

$$\mathbb{E}_D[\text{gap}] \leq \sqrt{\frac{R^*_{\mathcal{H}}\left(\log(4N_{2n}(\mathcal{H})) + 1\right)}{n}} + 20\, \frac{\log(4N_{2n}(\mathcal{H})) + 1}{n}.$$

Asymptotically,

$$\mathbb{E}_D\left[R(\hat{h}) - \hat{R}(\hat{h})\right] \leq O\left(\sqrt{\frac{(A_{\mathcal{H}} + \sigma^2)\, \log N_{2n}(\mathcal{H})}{n}} + \frac{\log N_{2n}(\mathcal{H})}{n}\right)$$

because, at the numerator,

$$\log(4N_{2n}(\mathcal{H})) + 1 = \log 4 + \log N_{2n}(\mathcal{H}) + 1 = \log N_{2n}(\mathcal{H}) + O(1)$$

and for large x,

$$\log x + O(1) \leq C \log x$$

for some constant $C > 0$. Plugging in the counting inequality from Lemma 2.4 with $2n$, we obtain and plot the following bound

$$\mathbb{E}_D\left[R(\hat{h}) - \hat{R}(\hat{h})\right] \leq \min\left\{O\left(\sqrt{\frac{(A_{\mathcal{H}} + \sigma^2)\log|\mathcal{H}|}{n}} + \frac{\log|\mathcal{H}|}{n}\right), O\left(\sqrt{A_{\mathcal{H}} + \sigma^2} + 1\right)\right\}$$

Remark: recall that $N_{2n}(\mathcal{H})$ is the maximum number of loss vectors over the union of two datasets D, D', or equivalently over a dataset with size $2n$. The counting lemma must be applied to $2n$-long loss vectors. Hence, we still have $N_{2n}(\mathcal{H}) \leq |\mathcal{H}|$ because the number of functions is still the same, and the bound counting all the possible sequences becomes $N_{2n}(\mathcal{H}) \leq 2^{2n}$.

□

Proof of Lemma 4.2. $N_D(\mathcal{H}) \leq \Pi_n(\mathcal{H})$ because the number of distinct loss vector can at most decrease, given prediction vectors. $\Pi_n(\mathcal{H}) \leq |\mathcal{H}|$ because the realizations of functions in h are at most as much as the functions in h (equality when functions in h are bijective). $\Pi_n(\mathcal{H}) \leq 2^n$ because we have at most 2^n sequences of n 1s and -1s. □